

Carlos Gafa'

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EDUCATION

University of Malta

Micro-credential in Intelligent Algorithmic Trading (ARI5123) Feb 2025 – Grade Pending

- Project: Determining Entry/Exit Points of a Pairs Trading Strategy Using Online Reinforcement Learning
- Relevant Coursework: Reinforcement Learning (DQN, Policy Gradient, Actor-Critic Methods), Hidden Markov Models, Technical Analysis

Micro-credential in Financial Engineering (ARI5122)

Nov 2024 – Feb 2025

- Grade: A
- Relevant Coursework: PCA, Volatility Models (GARCH, GBM, RNNs), Factor Investing

Master of Science in Engineering (Electrical) (By Research)

Oct 2023 – Jun 2025

- Passed with Distinction
- In collaboration with the Paul Scherrer Institute and the University of Cambridge
- Relevant Skills: Optimization Algorithms, Inverse Analysis Methods, Data Analysis and Statistics

Bachelor of Science (Hons.) in Mathematics and Physics

Sep 2019 – Jul 2023

- First Class Honors
- Dean's List of Science
- Relevant Coursework: Python Simulations, Markov Chain Monte Carlo Methods, Probability

G.F. Abela – Junior College

Matriculation Certificate

Sep 2017 – Jul 2019

- A' Levels in: Mathematics (A), Biology (A), Chemistry (A)

WORK EXPERIENCE

PricewaterhouseCoopers

Malta

Data Analyst

Nov 2024 – Ongoing

Key relevant experience:

- **Data Modelling:** Developed, and optimized data models and robust ETL pipelines for data warehousing initiatives to enhance data quality and reporting performance.
- **Report Development:** Created insightful reports using tools such as Power BI to support strategic, data-driven decisions.
- **Automation Tools:** Development of technology driven tools to automate financial risk calculations.
- **Technologies Used:** Python, SQL, Azure Data Factory, Power BI, Alteryx

Paul Scherrer Institute

Switzerland

Physicist

Sep 2023 – Nov 2024

During my time with the Insertion Devices group at PSI, I played a key role in the High-Temperature Superconducting Undulator (HTSU) project, wherein:

- I was involved in FEM simulations,
- I developed a shimming algorithm to make iterative geometrical design changes targeted to decrease the phase error of the HTSU,
- I developed gradient based algorithms to optimize the magnetic field integrals of the HTSU,

- I applied advanced statistical analysis and inverse analysis methods to identify and characterize key HTS parameters, and predict time-series decay (flux creep) in superconductors. Predictive models were validated against experimental data.

Additionally, I spent three weeks at the University of Cambridge, where I assisted in upgrading the HTSU's measurement system.

My work was accepted as an oral contribution at the HTS Modelling Workshop 2024 and Swiss Physical Society Meeting 2024.

PricewaterhouseCoopers

Data Analyst

Malta

Jun 2022 – Sep 2023 (Seasonal)

Peer Reviewed Publications

White Dwarf envelopes and temperature corrections in exponential $f(T)$ gravity

General Relativity and Gravitation

Dec 24, 2024

DOI: [10.1007/s10714-024-03341-6](https://doi.org/10.1007/s10714-024-03341-6)

Experimental results of a YBCO bulk superconducting undulator magnetic optimization

Physical Review Accelerators and Beams

Oct 10, 2024

DOI: [10.1103/PhysRevAccelBeams.27.100702](https://doi.org/10.1103/PhysRevAccelBeams.27.100702)

Professional Certifications

Microsoft

Microsoft Certified: Azure Data Fundamentals

Nov 2024

Credential ID: [D9FE60CD73CB9710](https://certification.microsoft.com/en-us/credential/D9FE60CD73CB9710)